Ceferino Patino

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Software engineer with hands-on experience building enterprise applications at Mutual of Omaha — developing RESTful APIs and backend services in Java and Groovy, building Angular and TypeScript front ends, and owning Python deployment orchestration and CI pipelines. Pairs production delivery with research-grade systems work, including a Rust+CUDA backend that doubled the performance of an international multi-agent reinforcement learning benchmark. Seeking full-time roles focused on full-stack development and developer tooling.

Education

Bachelor of Science (B.S.) in Computer Science — Lincoln, NE, United States
University of Nebraska – Lincoln — *May 2024 – May 2026*

Honors and Awards

Undergraduate Student Researcher Award — University of Nebraska – Lincoln, *2025*

Dean's List — University of Nebraska – Lincoln, *Fall 2024 – Spring 2026*

Skills

Languages: Python, C++, Rust, JavaScript, TypeScript, Groovy, Lua, Bash, SQL, Nix, Java

Frameworks & Libraries: PyTorch, OpenCV, Astro, React, Angular, Express.js

Tools & Platforms: Docker, Kubernetes (K8s), Terraform, Git, Nix/NixOS, Maturin, MPI, HDF5, AWS, Django, PostgreSQL, MySQL, Oracle SQL

Concepts: Machine Learning, RL/MARL

Software Engineering: SOLID, Agile, Scrum, CI/CD, DevOps, Parallel Programming

Soft Skills: Team collaboration, Technical communication, Self-directed learning, Mentorship

Experience

Mutual of Omaha — Remote
Application Developer – Java, Groovy, Angular, TypeScript, Python, Oracle SQL — *August 2026 – October 2026*

- Drove a previously failed two-year deployment to successful release by mirroring production traffic via nginx into a test environment and diffing payloads and stack traces, surfacing 50+ breaking bugs in two weeks
- Continued to own and extend Python deployment orchestration, CI pipelines, and cross-team tooling while taking on new application development beyond internship scope, further cutting CI build/scan cycles from 20 to 10 minutes
- Developed and maintained enterprise applications in Java and Groovy, building RESTful APIs and backend services supporting business operations
- Spearheaded a multi-stage modernization of the production codebase, upgrading Java from 8 to 17 to 21, Groovy from 2 to 3, and Spring Boot from 2.5.4 to 4.1.0 while keeping services stable and regression-free throughout the migration
- Built responsive front-end interfaces with Angular, JavaScript, and TypeScript for internal stakeholders
- Stabilized overnight Oracle SQL reporting and batch jobs (6 PM–6 AM window), cutting incidents from 120 to 17 per four months (about 86%) by surfacing failures at 10–11 PM instead of 2–3 AM so jobs complete by midnight

Mutual of Omaha — Remote
I/S Intern – Python, CI/CD, DevOps Tooling — *May 2026 – July 2026*

- Replaced a 12-page manual deployment runbook spanning 6–10 machines across 3 test environments and production with a Python orchestration tool, turning a recurring two-week quarterly task into a single automated run and eliminating configuration drift
- Built and maintained CI pipelines that automated build, test, and release validation, cutting after-hours deployment from 60–90 minutes to under 5 minutes and build/scan cycles from 30 to 20 minutes
- Partnered with peer teams to identify shared pain points and distribute reusable tooling and solutions, standardizing delivery practices across groups

OASYS Lab – University of Nebraska – Lincoln — Lincoln, NE, United States
Undergraduate Researcher – Python, PyTorch, Rust, CUDA, Maturin — *May 2024 – August 2026*

- Developed `free-range-zoo`, an open-environment MARL benchmark suite (wildfire, cybersecurity, rideshare) used by 20+ researchers at international institutions
- Built `free-range-rust`, a Rust+CUDA backend with Maturin/Python bindings, doubling RL environment throughput over the Python implementation
- Implemented dynamic agent/task/frame adaptation enabling open-system evaluation of MARL policies
- Co-led MOASEI at AAMAS 2025, an open-agent/task MARL competition with 10+ international teams; co-authored the technical report (arXiv:2507.05469)

Raman Lab – Washington University — Remote
Software Developer – C++, MPI, HDF5, OpenCV — *June 2024 – May 2026*

- Developed real-time imaging software for a 3D 2-photon lightsheet microscope processing 20 GB/s of image data
- Designed parallel MPI/HDF5 data acquisition and OpenCV pipelines for volumetric dataset processing

CSCE361 Software Engineering – University of Nebraska – Lincoln Lincoln, NE, United States
Undergraduate TA – C#, .NET, React, Node.js, Agile *August 2025 – May 2026*

- Supported 30–60 students per semester in labs, assignments, and team projects for CSCE361, a course on large-scale software development and engineering best practices
- Taught the course technology stack and software engineering principles through weekly walkthroughs and office hours, covering C#, .NET Framework, EF Core, React, Node.js, JavaScript, TypeScript, SOLID design, and Scrum
- Graded quizzes, exams, and project deliverables, providing feedback on requirements analysis, design, testing, and maintenance

Danforth Plant Science Center St. Louis, MO, United States
Software Developer – Python, OpenCV, PyTorch, AWS, Docker *December 2023 – August 2024*

- Developed computer vision software for plant phenotyping, segmenting high-resolution underground root imagery to compute volumetric root estimates at 97% accuracy
- Maintained AWS infrastructure with developer teams processing 40 GB hyperspectral aerial crop imagery campaigns captured at 10 fps from a high-throughput drone data collection system

Open Source Contributions

nixpkgs NixOS Project
Contributor & Package Maintainer *May 2025 – Present*

- Landed 50+ merged PRs across nixpkgs, spanning new packages, version updates, and code reviews
- Review community contributions and validate builds to keep packages reproducible across supported platforms

nixvim NixOS Project
Maintainer *May 2025 – Present*

- Maintain Nixvim, a declarative Neovim configuration framework for Nix, reviewing community PRs for plugin modules and documentation
- Add and update plugin support for popular Vim/Neovim integrations and improve community-facing docs

Turso Distributed Edge Database
Contributor & Package Maintainer *2025*

- Fixed a core reliability bug in Turso, a distributed edge database built on libSQL, improving correctness for production users
- Collaborated with upstream maintainers to review, test, and merge the fix into the main codebase

Projects

cpatino.com Personal
Web Development – JavaScript, Astro *May 2025 – Present*

- Built and deployed a fully static personal site with Astro, shipping near-zero client-side JavaScript for fast page loads

yumevim – "dream vim" Personal
Dev Tooling – Nix, Lua, Neovim, Nixvim *July 2024 – Present*

- Architected a modular, declarative Neovim configuration in Nix, packaged as a Nix flake that runs with a single command (`nix run`) on any machine
- Built minimal and full profiles with LSP, Treesitter, DAP, and formatting via Nixvim, enforcing consistent code style with CI and Alejandra formatting
- Replaced a hand-maintained dotfile with a reproducible package, giving zero-downtime updates and atomic rollbacks through Nix flakes and Git

yumeami – "dream network" Personal
DevOps – Nix, NixOS *April 2024 – Present*

- Engineered a 6-machine heterogeneous NixOS fleet (x86 and ARM) with a single flake, unifying system and home configurations across desktops, servers, a laptop, and a Raspberry Pi
- Automated provisioning end to end with disko and nixos-anywhere, installing fresh machines remotely over SSH with encrypted git-crypt secrets and self-hosted services backed by PostgreSQL and MySQL
- Implemented ZFS-impermanence for atomic rollbacks on every boot, keeping 145+ shared modules consistent and reproducible across the whole fleet

3D Volumetric 2-Photon Lightsheet Microscope Raman Lab – Washington University
Biotechnology / Microscopy – C++, MPI, HDF5, OpenCV *June 2024 – August 2026*

- Developed software for a cutting-edge 3D volumetric 2-photon lightsheet microscope—one of few worldwide
- Built parallel data acquisition pipelines with MPI and HDF5 handling 20 GB/s of image data
- Optimized imaging workflows with OpenCV for real-time processing of volumetric datasets

free-range-zoo OASYS Lab – University of Nebraska – Lincoln
AI / Reinforcement Learning – Python, PyTorch *April 2024 – August 2026*

- Developed open-environment benchmarks for POSG-based multi-agent RL (wildfire, cybersecurity, rideshare), used internationally by 20+ researchers
- Implemented vectorized training loops for MADDPG, COMA, and GNN-based RL policies
- Enabled dynamic agent/task/frame changes to benchmark policy adaptability in open systems

free-range-rust OASYS Lab – University of Nebraska – Lincoln
AI / Reinforcement Learning – Rust, CUDA, Maturin *April 2024 – August 2026*

- Implemented custom CUDA kernels in Rust to accelerate dynamic RL environments

- Doubled performance of `free-range-zoo` through low-level vectorized space operations
- Exposed Rust/CUDA modules to Python via Maturin for seamless integration and use by other research groups

UNL Senior Design Capstone (Mutual of Omaha – Sponsor) Lincoln, NE, United States
Student Software Engineer *September 2025 – May 2026*

- Co-led a 5–6 person team designing and developing a fullstack JSON validation engine for Mutual of Omaha internal workflow automation
- Built and deployed the solution using React (frontend), Spring Boot (API backend), and ECS/Aurora for production cloud delivery, provisioning infrastructure with Terraform before migrating to AWS CDK
- Collaborated with Mutual of Omaha engineers to deliver requirements, review code, and build CI/CD pipelines for cloud deployment

Fonio Seed Computer Vision Donald Danforth Plant Science Center
Computer Vision – Roboflow, OpenCV *April 2024 – August 2024*

- Compiled and labeled a dataset of over 8,000 fonio seed images for robust model training
- Implemented a neural network achieving 99% accuracy in detecting overlapping seeds and providing precise counts

SubTerra.ai Donald Danforth Plant Science Center
Computer Vision / Agricultural AI – PyTorch, OpenCV, Django, Docker *December 2023 – August 2024*

- Developed a Django interface to run segmentation and extrapolate root volume
- Built root segmentation models using PyTorch for high-resolution root image data
- Processed segmentation masks with OpenCV to compute volumetric estimates achieving 97% accuracy

FieldDock Donald Danforth Plant Science Center
Computer Vision / Agricultural AI – OpenCV, Docker, AWS *December 2023 – May 2024*

- Containerized and deployed components of a high-throughput drone data collection system using Docker and AWS, processing multi-gigabyte aerial imagery campaigns
- Developed flight software for drone vision and landing leveraging OpenCV

qOverflow Black Data Processing Associates (BDPA)
Web Development – JavaScript, MongoDB, React, Express.js, Node.js *June 2022 – August 2022*

- Won first place in 2022 BDPA hackathon developing a Stack Overflow-inspired Q&A platform with custom analytics
- Containerized the app using Docker and deployed on Kubernetes; documented APIs with SwaggerHub

hypixel-helper Personal
Web Development – JavaScript, React, Redis *April 2022 – May 2022*

- Built live tracking and forecasting of Hypixel Skyblock marketplace data using SARIMA

Publications

Second MOASEI Competition at AAMAS'2026: A Technical Report arXiv:2607.03399
Patino C., Billings T., Abadi A., Redder D., Eck A., Doshi P., Soh L. *2026*

Inaugural MOASEI Competition at AAMAS'2025: A Technical Report arXiv:2507.05469
Patino C., Billings T., Abadi A., Redder D., Eck A., Doshi P., Soh L. *2025*